

Faiq Shahbaz

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RESEARCH INTERESTS

My research interests lie in computational fluid dynamics, turbulence modelling, numerical methods, high-performance scientific computing, and scientific machine learning for fluid mechanics. I am particularly interested in developing reliable and physically consistent computational and data-driven methods that improve the accuracy, efficiency, and predictive capability of CFD.

EDUCATION

National University of Sciences & Technology (NUST)

Master of Science in Computational Science & Engineering

Islamabad, Pakistan

Sep. 2023 – Aug. 2025

- **CGPA:** 3.60/4.00 — **Thesis Grade:** A
- **Thesis:** *CFD-Based Design and Optimization of a Ducted Wind Turbine* (Supervisor: Dr. Ammar Mushtaq)
- **Relevant Coursework:** Advanced Fluid Mechanics, CFD-I/II, High-Performance Computing, Computational Linear Algebra & Optimization, Applied Mathematics.

Institute of Space Technology (IST)

Bachelor of Science in Aerospace Engineering

Islamabad, Pakistan

Sep. 2018 – Jul. 2022

- **Focus:** Computational Fluid Dynamics and Aerodynamics
- **Thesis:** *Design & Analysis of an Airborne Wind Energy System*

EXPERIENCE

CFD Support Engineer

BetaCodes (Formerly Forbmax)

May 2025 – Present

Islamabad, Pakistan

- Work under the supervision of Dr. Rooh Khurram on CFD research and HPC activities involving turbulence modelling, data-driven CFD, and scientific computing.
- Develop CFD training material on turbulence modelling, meshing, numerical methods, and data-assisted CFD for HPC outreach and education activities with KAUST.
- Support academic and industrial users of KAUST's Shaheen supercomputer with case preparation, numerical troubleshooting, solver performance, and parallel CFD workflows.
- Support cloud-based HPC clusters used for CFD training and simulation, including workload execution, parallel scaling, and HPC workflow guidance.

Research Associate

Computational Aeronautics Lab (CAL), NUST

Jul. 2024 – May 2025

Islamabad, Pakistan

- Conducted funded MSc thesis research on the design and optimization of a diffuser-augmented micro wind turbine using low-Reynolds-number airfoil screening, BEMT, and blade-resolved steady RANS simulations (SST $k-\omega$, MRF).
- Developed a Python-based BEMT solver and evaluated multiple rotor-diffuser configurations; CFD predicted an increase in peak rotor-area-normalized power coefficient from 0.43–0.44 for the bare rotor to 0.86–0.89 with diffuser augmentation.
- Conducted prototype fabrication and blower-based experiments, with the ducted turbine producing $1.38\text{--}2.41\times$ the electrical power of the bare turbine over their common operating range.
- Managed the CAL computing cluster and supported CFD simulations for ongoing aerodynamics research projects.

Computational Science Intern

Dense Fusion

Jul. 2024 – Oct. 2024

Islamabad, Pakistan

- Implemented PETSc-based solvers for canonical PDEs, including the Poisson equation, and worked with OpenFOAM utilities for numerical simulation of transport-equation problems.
- Developed basic MPI-parallel implementations and performed scalability tests on multi-core systems, alongside Python automation for simulation and CFD dataset generation.
- Collaborated with KAUST researchers in preparing computational-methods training material for CFD and numerical modelling workshops.

RESEARCH PROJECTS

Ahmed Body Turbulence Modelling, Surrogate Modelling & Digital Twin

Tools: ANSYS Fluent, Python, scikit-learn, SALib, XGBoost, PyTorch Geometric

- Performed a parametric GEKO-RANS study of the 25° Ahmed body, including mesh sensitivity assessment and validation against published force and wake measurements; identified the GEKO separation parameter (C_{SEP}) as the dominant parameter and observed a sharp transition in lift behaviour.
- Developed and compared Gaussian process, radial basis function, MLP, and XGBoost surrogate models across multiple inlet velocities, with sensitivity analysis, uncertainty quantification, and reliability assessment for drag and lift prediction.
- Extended the study using LES and DES simulations for high-fidelity comparison of aerodynamic forces and separated wake behaviour with the RANS predictions.
- Currently extending the framework from coefficient-level surrogate prediction toward a digital twin using graph neural networks to predict spatial pressure and velocity fields.

Reynolds-Conditioned Graph Neural Networks for Cavity Flow Prediction

Tools: Python, PyTorch, PyTorch Geometric, HDF5

- Developed an end-to-end graph neural network framework for parametric cavity flow prediction using CFD datasets spanning $Re = 100$ – $10,000$, including a 100-case dataset on a computational mesh with one million cells.
- Constructed graph representations directly from CFD mesh connectivity, with Reynolds-number conditioning and physically consistent nondimensionalization for prediction of velocity and pressure fields.
- Implemented six GNN architectures, including GCN, GraphSAGE, GATv2, GINE, NNConv, and Graph U-Net, using leakage-free protocols for interpolation across Reynolds numbers and extrapolation to higher Reynolds numbers.
- Built reproducible training and evaluation workflows with physical-space error metrics, benchmark comparisons, automated validation, checkpointing, and Slurm-based HPC experiment configurations.

Graph Neural Network Digital Twin for Airfoil Aerodynamics

Tools: OpenFOAM, Python, PyTorch, PyTorch Geometric, Streamlit

- Developed a digital twin framework for NACA0012 airfoil flow prediction by combining validated CFD simulations with graph neural network surrogate modelling over angles of attack from -4° to 16° and Reynolds numbers from 3×10^6 to 9×10^6 .
- Constructed graph datasets directly from finite-volume mesh connectivity, using geometry, boundary information, Reynolds number, and angle of attack as deployable inputs for prediction of velocity, pressure, and turbulence-model fields.
- Designed reproducible training, validation, and test protocols to evaluate generalization across unseen angles of attack and Reynolds numbers, including interpolation and extrapolation conditions.
- Developed a runtime inference pipeline and Streamlit dashboard that prevent data leakage by using only geometry, mesh topology, Reynolds number, and angle of attack to generate and visualize predicted flow fields for unseen operating conditions.

MANUSCRIPTS & CONFERENCE SUBMISSIONS

F. Shahbaz, H. Shafiq, Q. S. Ali, A. Mushtaq. “*Aerodynamic Design, CFD Analysis, and Experimental Power Assessment of a Diffuser-Augmented Micro-Scale Horizontal-Axis Wind Turbine for Low-Wind-Speed Environments.*” Submitted to *Wind* (MDPI).

F. Shahbaz, R. Khurram. “*Sensitivity of GEKO Turbulence-Model Parameters for the 25° Ahmed Body.*” Submitted to the *24th IACM Computational Fluids Conference (CFC 2027)*, Yokohama, Japan.

F. Shahbaz, R. Khurram. “*Reliability-Aware Surrogate Modelling of GEKO-RANS Aerodynamics for the 25° Ahmed Body.*” Submitted to the *4th IACM Digital Twins in Engineering Conference (DTE 2027)*, Yokohama, Japan.

SKILLS & TOOLS

Programming: Python, C++, MATLAB, Bash

HPC & Numerical Computing: MPI, PETSc, Slurm, Linux, Git, CMake, Docker, AWS

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn

CFD & Solvers: OpenFOAM, DAFoam, ANSYS Fluent, STAR-CCM+

Meshing & CAD: Pointwise, ICEM CFD, CATIA, Blender

Scientific Tools: NumPy, SciPy, HDF5, ParaView, Matplotlib, LaTeX